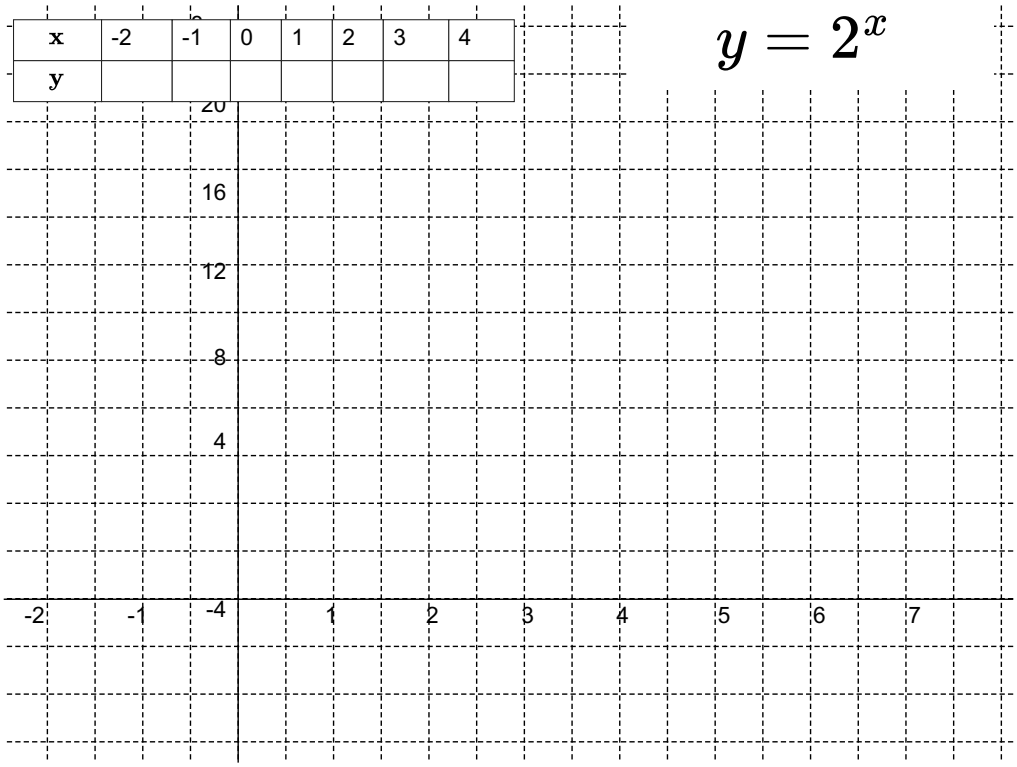
1.22 Exponentials and Logarithms

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x	-2	-1	0	1	2	3	4
y							

$y = 2^x$

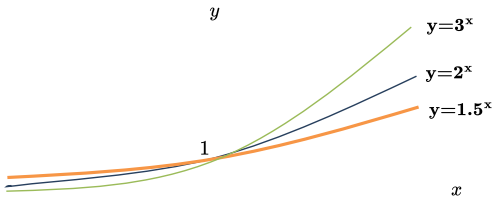


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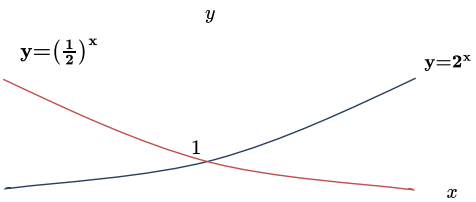
Contrasting exponential graphs

5

On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$



On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$



Three important notes:

- $y = 2^x$ is said to be "**exponential growing**" whereas $y = \left(\frac{1}{2}\right)^x$ is said to be "**exponentially decaying**", because it's getting smaller (halving) each time x increases by 1.
- $y = \left(\frac{1}{2}\right)^x$ is a reflection of $y = 2^x$ in the line $x = 0$.
  Proof:
  If $f(x) = 2^x$, $f(-x) = 2^{-x} = \frac{1}{2^x} = \left(\frac{1}{2}\right)^x$
- $\left(\frac{1}{2}\right)^x$ would usually be written 2^{-x}.

You should therefore in general be able to recognise and sketch the graph $y = a^{-x}$.

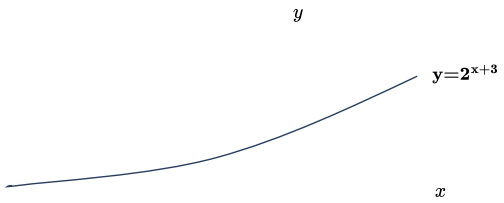
1.22 Exponentials and Logarithms

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Graph Transformations

6

Sketch $y = 2^{x+3}$



The 'change' to 2^x is 'inside the function' (i.e. input x is replaced with $x + 3$).

Ensure you work out the new y-intercept.

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Exercise 14A (Pages 313-314)

7

6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

a By drawing a sketch or otherwise, explain why $0 < q < 1$.

b Find the values of the constants p and q.

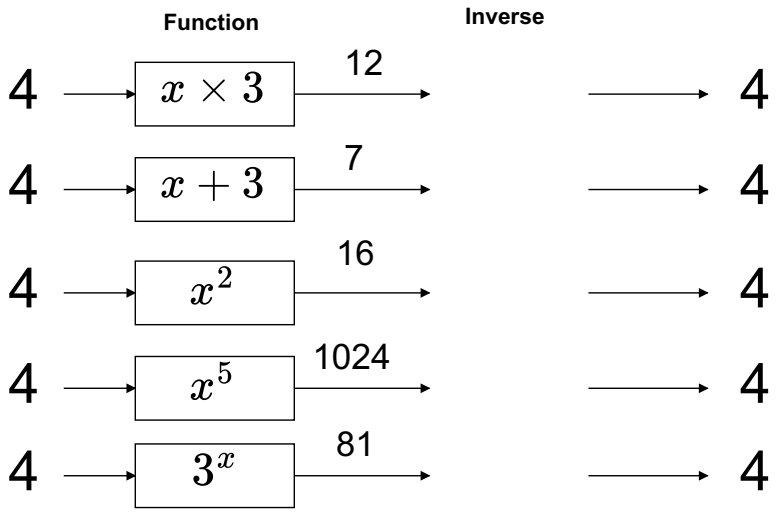
1.22 Exponentials and Logarithms

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Logarithms

21

You know the inverse of many mathematical operations; we can undo an addition by 2 for example by subtracting 2. But is there an inverse function for an exponential function?



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Interchanging between exponential and log form

22

$\log_a n$ ("said log base a of n") is equivalent to $a^x = n$.
The log function outputs the **missing power**.

$$3^2 = 9 \iff \log_3 9 = 2$$

Here are two methods of interchanging between these forms.
Pick your favourite!

Method 1: 'Missing Power'

- Note first the base of the log must match the base of the exponential.
- $\log_2 8$ for example asks the question "2 to **what power** gives 8?"

We can imagine inserting the output of the log just after the base. Click the button!

$\log_{\quad 2} 8 = 3$

Click to start Animation

Method 2: Do same operation to each side of equation.

Since KS3 you're used to the idea of doing the same thing to each side of the equation that 'undoes' whatever you want to get rid of.

$$3x + 2 = 11$$
$$\overset{(-2)}{3x} = 9^{(-2)}$$

"log base a" undoes "a to the power of" and vice versa, as they are inverse functions.

$$\log_2 8 = 3$$
$$\overset{(2^{\square})}{8} = 2^{3^{(2^{\square})}}$$

1.22 Exponentials and Logarithms

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Examples

23

$\log_5 25 =$

Think: "5 to the power of what gives you 25?"

$\log_3 81 =$

$\log_2 32 =$

$\log_{10} 1000 =$

$\log_4 1 =$

$\log_4 4 =$

$\log_2(\frac{1}{2}) =$

$\log_3(\frac{1}{27}) =$

$\log_2(\frac{1}{16}) =$

$\log_a(a^3) =$

$\log_4(-1) =$

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With Your Calculator

24

There are three buttons on your calculator for computing logs:

$\log_{\square}\square$

$\log_3 7 =$

$\log_5 0.3 =$

ln

$\ln 10 =$

$\ln e =$

log

$\log 100 =$

Just like the $\sqrt{}$ symbol without a number is $\sqrt[2]{\square}$ by default, $\log$ without a base is **base 10** by default when used on your calculator (although confusingly "log" can mean "ln" in mathematical papers)

$\ln$ is the "**natural log of x**", meaning "log to the base e", i.e. it the inverse of e^x.

$$\ln(x) = \log_e(x)$$

We will use it more extensively later this chapter.

1.22 Exponentials and Logarithms

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Exercise 14D – Y1 Page 321

25

6 a Without using a calculator, justify why the value of $\log_2 50$ must be between 5 and 6.

b Use a calculator to find the exact value of $\log_2 50$ to 4 significant figures.

7 a Find the values of:

i $\log_2 2$ **ii** $\log_3 3$ **iii** $\log_{17} 17$

b Explain why $\log_a a$ has the same value for all positive values of a ($a \neq 1$).

8 a Find the values of:

i $\log_2 1$ \qquad **ii** $\log_3 1$ \qquad **iii** $\log_{17} 1$

b Explain why $\log_a 1$ has the same value for all positive values of a ($a \neq 1$).

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Extension Exercise

26

1 [MAT 2015 1J] Which is the largest of the following numbers?

A) $\frac{\sqrt{7}}{2}$ B) $\frac{5}{4}$ C) $\frac{\sqrt{}}{3(}$ Non-calculator!

D) $\frac{\log_2 30}{\log_3 85}$ E) $\frac{1+\sqrt{6}}{3}$

2 [MAT 2013 1F] Three positive numbers a, b, c satisfy

$\log_b a = 2$

$\log_b(c-3) = 3$

$\log_a(c+5) = 2$

This information:

specifies a uniquely;

is satisfied by two values of a;

is satisfied by infinitely many values of a;

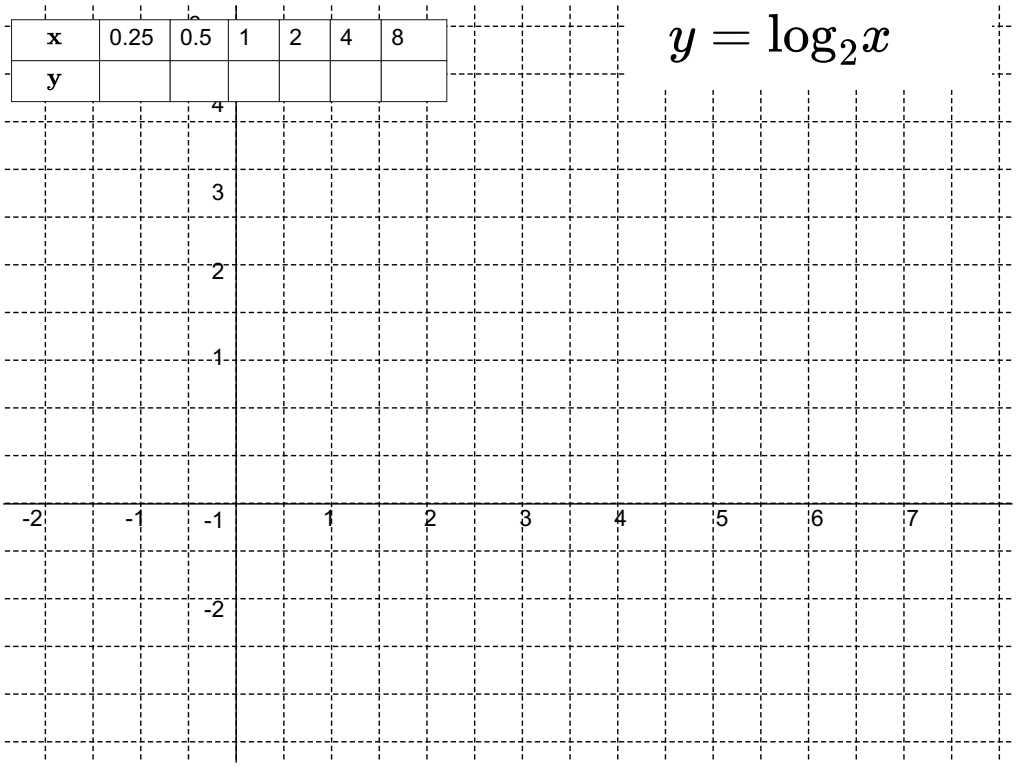
is contradictory

1.22 Exponentials and Logarithms

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x	0.25	0.5	1	2	4	8
y						

$y = \log_2 x$



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Laws of Logs

28

Three main laws:

$\log_a x + \log_a y = \log_a xy$

$\log_a x - \log_a y = \log_a \left(\frac{x}{y}\right)$

$\log_a \left(x^k\right) = k\log_a x$

The logs must have a consistent base.

i.e. You can move the power to the front.

Special cases:

$\log_a a = 1 \quad (a > 0, a \neq 1)$

$\log_a 1 = 0 \quad (a > 0, a \neq 1)$

$\log\left(\frac{1}{x}\right) = \log\left(x^{-1}\right) = -\log(x)$

We often try to avoid leaving fractions inside logs.
So if the answer was:

$\log_2\left(\frac{1}{3}\right)$

You should write your answer as: $-\log_2 3$

Reciprocating the input negates the output.

Not in syllabus (but in MAT/PAT):

$$\log_a b = \frac{\log_c b}{\log_c a}$$

This is known as **changing the base**.
So to get $\log_2 9$ in terms of log base 3:

$log_2 9 = \frac{log_3 9}{log_3 2} = \frac{2}{log_3 2}$

1.22 Exponentials and Logarithms

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Examples

29

Write as a single logarithm:
a. $\log_3 6 + \log_3 7$
b. $\log_2 15 - \log_2 3$
c. $2\log_5 3 + 3\log_5 2$
d. $\log_{10} 3 - 4\log_{10}\left(\frac{1}{2}\right)$

Write in terms of $\log_a x$, $\log_a y$ and $\log_a z$
a. $\log_a(x^2yz^3)$
b. $\log_a\left(\frac{x}{y^3}\right)$
c. $\log_a\left(\frac{x\sqrt{y}}{z}\right)$
d. $\log_a\left(\frac{x}{a^4}\right)$

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Anti Laws

30

These are **NOT LAWS OF LOGS**, but are mistakes students often make:

$$\log_a(b+c) = \log_a b + \log_a c$$

There is **no method** to simplify the **log of a sum**, only the sum of two logs!

✗ FAIL

$$(\log_2 x)^3 = 3\log_2 x$$

The power must be on the input (here the x), but here the power is around the entire log.

✗ FAIL

1.22 Exponentials and Logarithms

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Solving Equations with Logs

31

Solve the equation $\log_{10}4 + 2\log_{10}x = 2$

This is a very common type of exam question.
The strategy is to **combine the logs into one** and isolate on one side.

We've used the laws of logs to combine them into one.

Use your favourite method of rearranging. Either do "10 the power of each side" to "undo" the log, or the "insert the 2 between the 10 and $4x^{2}$" method.

The subtle bit: You must check each value in the original equation. We're not allowed to log a negative number.

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Test Your Understanding

32

Edexcel C2 Jan 2013 Q6

Given that $2\log_2(x+15) - \log_2 x = 6$,

(a) show that $x^2 - 34x + 225 = 0$. \hfill (5)

(b) Hence, or otherwise, solve the equation $2\log_2(x+15) - \log_2 x = 6$. \hfill (2)

a

Those who feel confident with their laws could always skip straight to this line.

b

1.22 Exponentials and Logarithms

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Exercise 14E (323-324)

33

5 a Given that $\log_3(x+1) = 1 + 2\log_3(x-1)$, show that $3x^2 - 7x + 2 = 0$.

b Hence, or otherwise, solve $\log_3(x+1) = 1 + 2\log_3(x-1)$.

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Exercise 14E (323-324)

34

6 Given that a and b are positive constants, and that $a > b$, solve the simultaneous equations

$a + b = 13$

$\log_6 a + \log_6 b = 2$

Challenge:
By Writing $\log_a x = m$ and $\log_a y = n$
Prove that
$\log_a x - \log_a y = \log_a \left(\frac{x}{y}\right)$

1.22 Exponentials and Logarithms

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Extension Exercises

35

These are all strictly **non-calculator**!

1 [AEA 2010 Q1b] Solve the equation

$$\log_3(x-7) - \tfrac{1}{2}\log_3 x = 1 - \log_3 2$$

2 [AEA 2008 Q5i] Anna, who is confused about the rules of logarithms, states that

$$(\log_3 p)^2 = \log_3(p^2)$$

$$\log_3(p+q) = \log_3 p + \log_3 q$$

However, there is a value for p and a value for q for which both statements are correct. Find their values.

3 [MAT 2007 1I] Given that a and b are positive and

$$4(\log_{10} a)^2 + (\log_{10} b)^2 = 1$$

what is the greatest possible value of a?

4 [MAT 2002 1F] Observe that $2^3 = 8$, $2^5 = 32$, $3^2 = 9$ and $3^3 = 27$. From these facts, we can deduce that $\log_2 3$ is:

between $1\frac{1}{3}$ and $1\frac{1}{2}$

between $1\frac{1}{2}$ and $1\frac{2}{3}$

between $1\frac{2}{3}$ and 2

none of the above

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Solving equations with exponential terms

36

Solve $3^x = 20$

This is often said “Taking logs of both sides…”

Solve $5^{4x-1} = 61$

1.22 Exponentials and Logarithms

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Solving equations with exponential terms

37

Solve $3^x = 2^{x+1}$

Why can we not apply quite the same strategy here?

Logs in general are great for solving equations when the variable is in the power, because laws of logs allow us to move the power down.

This then becomes a GCSE-style 'changing the subject' type question. Just isolate x on one side and factorise out.

It doesn't matter what base you use to get the final answer as a decimal, provided that it's consistent. You may as well use the calculator's 'log' (no base) key.

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Test Your Understanding

38

1 Solve $3^{2x-1} = 5$, giving your answer to 3dp.

2 Solve $3^{x+1} = 4^{x-1}$, giving your answer to 3dp.

3 Solve $2^x 3^{x+1} = 5$, giving your answer in exact form.

1.22 Exponentials and Logarithms

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Exercise 14F – Y1 Page 325

39

3 Solve the following equations, giving your answers to 3 significant figures where appropriate.

a $3^{x+1} = 2000$ \hfill (2 mar

b $\log_5(x-3) = -1$ \hfill (2 mar

a $3^{x+1} = 2000$ **b** $5^{-1} = x - 3$

4 a Sketch the graph of $y = 4^x$, stating the coordinates of any points where the graph crosses the axes.

b Solve the equation $4^{2x} - 10(4^x) + 16 = 0$.

1 EXTENSION [MAT 2011 1H] How many positive values x which satisfy the equation:

$$x = 8^{\log_2 x} - 9^{\log_3 x} - 4^{\log_2 x} + \log_{0.5} 0.25$$

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Very Challenging Extension!

40

2 [MAT 2013 1J] For a real number x we denote by $[x]$ the largest integer less than or equal to x. Let n be a natural number. The integral

$$\int_0^n [2^x] dx$$

equals:
(A) $\log_2((2^n - 1)!)$
(B) $n2^n - \log_2((2^n)!)$
(C) $n2^n$
(D) $\log_2((2^n)!)$

